



AMITY UNIVERSITY PUNJAB, MOHALI

B.SC. MATHEMATICS (5TH SEMESTER)
SET THEORY AND METRIC SPACES (MAT304)

Unit II: Concepts in Metric Spaces

Complete Lecture Notes

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1 Introduction to Metric Spaces

1.1 The Central Idea: Abstracting Distance

In Euclidean geometry, we have an intuitive notion of distance between points. The brilliance of the metric space concept—developed by Maurice Fréchet in 1906—was to **abstract** this notion and ask:

What are the minimal properties a "distance" function must satisfy to be useful?

The answer turns out to be surprisingly simple: only four axioms are needed. We call such a distance function a **metric**.

1.2 Definition of a Metric Space

Definition 1.1 (Metric Space). Let X be a non-empty set. A function $d : X \times X \rightarrow [0, \infty)$ is called a **metric** (or **distance function**) if for all $x, y, z \in X$:

(i) **Non-negativity and Identity of Indiscernibles:**

$$d(x, y) = 0 \iff x = y.$$

(ii) **Symmetry:**

$$d(x, y) = d(y, x).$$

(iii) **Triangle Inequality:**

$$d(x, z) \leq d(x, y) + d(y, z).$$

The pair (X, d) is called a **metric space**.

Remark 1.1 (The Triangle Inequality Intuition). The triangle inequality captures the fundamental geometric truth that *the shortest path between two points is the direct one*. Any detour through a third point y cannot be shorter than going directly from x to z .

Remark 1.2 (Notation). If the metric is understood from context, we often refer to the underlying set X itself as the metric space.

1.3 Rich Examples of Metric Spaces

Example 1.1 (The Real Line — Euclidean Metric). The most familiar example: $X = \mathbb{R}$, with $d(x, y) = |x - y|$. This satisfies all four axioms:

- $d(x, y) = 0 \iff |x - y| = 0 \iff x = y.$
- $d(x, y) = |x - y| = |y - x| = d(y, x).$
- $d(x, z) = |x - z| \leq |x - y| + |y - z| = d(x, y) + d(y, z).$

Example 1.2 (Euclidean n -Space). For $X = \mathbb{R}^n$, with points $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$, define

$$d_2(x, y) = \left(\sum_{k=1}^n |x_k - y_k|^2 \right)^{1/2}.$$

This is the standard Euclidean distance. It is the metric you use when measuring straight-line distances in the plane or space.

Example 1.3 (The Taxicab Metric). On $X = \mathbb{R}^n$, define

$$d_1(x, y) = \sum_{k=1}^n |x_k - y_k|.$$

This is called the **taxicab** or **Manhattan** metric because it measures distance as if you were traveling along a grid of city streets.

Intuition: In Manhattan, the shortest path from point A to point B is not a straight line through buildings, but the sum of horizontal and vertical distances.

Example 1.4 (The Supremum Metric). On $X = \mathbb{R}^n$, define

$$d_\infty(x, y) = \max_{1 \leq k \leq n} |x_k - y_k|.$$

This is called the **supremum** or **chessboard** metric.

Intuition: The king in chess moves one square in any direction. The number of moves a king needs to go from one square to another is exactly the maximum of the horizontal and vertical differences.

Example 1.5 (The Discrete Metric). For **any** non-empty set X , define:

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

This is called the **discrete metric**. In this space, every point is isolated—all distinct points are at distance exactly 1 from each other.

Example 1.6 (Sequence Spaces — ℓ^1 , ℓ^2 , ℓ^∞). Let ℓ^1 be the set of all sequences $(x_n)_{n=1}^\infty$ of real numbers such that $\sum_{n=1}^\infty |x_n| < \infty$. Define

$$d_1(x, y) = \sum_{n=1}^\infty |x_n - y_n|.$$

Similarly, ℓ^2 is the set of sequences with $\sum |x_n|^2 < \infty$, with

$$d_2(x, y) = \left(\sum_{n=1}^\infty |x_n - y_n|^2 \right)^{1/2}.$$

And ℓ^∞ is the set of bounded sequences, with

$$d_\infty(x, y) = \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

Example 1.7 (Function Spaces — The Supremum Metric). Let $X = C([a, b])$ be the set of all continuous real-valued functions on the closed interval $[a, b]$. For $f, g \in X$, define:

$$d_\infty(f, g) = \sup_{a \leq x \leq b} |f(x) - g(x)|.$$

This is a metric space! The distance between two functions is the maximum vertical separation between their graphs.

Example 1.8 (Function Spaces — The L^1 Metric). On the same set $C([a, b])$, define:

$$d_1(f, g) = \int_a^b |f(x) - g(x)| dx.$$

This measures the area between the graphs of f and g .

Example 1.9 (Matrix Spaces — The Frobenius Metric). Let $M_{m \times n}(\mathbb{R})$ be the set of all $m \times n$ real matrices. Define the Frobenius metric:

$$d_F(A, B) = \left(\sum_{i=1}^m \sum_{j=1}^n |a_{ij} - b_{ij}|^2 \right)^{1/2}.$$

This is the Euclidean distance in the mn -dimensional space of matrices.

Alternatively, the operator norm metric:

$$d_{\text{op}}(A, B) = \sup_{\|x\| \leq 1} \|(A - B)x\|_2,$$

where $\|\cdot\|_2$ is the Euclidean norm in $\mathbb{R}^n$. This metric measures how much the linear transformations represented by A and B differ at worst.

1.4 The Unit Ball in $\mathbb{R}^2$ with Different Metrics

The **unit ball** centered at the origin is defined as:

$$B_1(0) = \{x \in \mathbb{R}^2 : d(x, 0) < 1\}.$$

Let us explicitly compute and visualize the unit ball for each of the three main metrics on $\mathbb{R}^2$.

(a) **Euclidean Metric** (d_2):

$$B_1(0) = \{(x, y) \in \mathbb{R}^2 : \sqrt{x^2 + y^2} < 1\}.$$

This is the **open disk** of radius 1 centered at the origin—a perfect circle.

(b) **Taxicab Metric** (d_1):

$$B_1(0) = \{(x, y) \in \mathbb{R}^2 : |x| + |y| < 1\}.$$

This is a **diamond** (a square rotated by 45°) with vertices at $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$.

Why? The equation $|x| + |y| = 1$ describes a square in the L^1 norm. In the first quadrant, this is $x + y = 1$, which is a line segment from $(1, 0)$ to $(0, 1)$.

(c) **Supremum Metric** (d_∞):

$$B_1(0) = \{(x, y) \in \mathbb{R}^2 : \max\{|x|, |y|\} < 1\}.$$

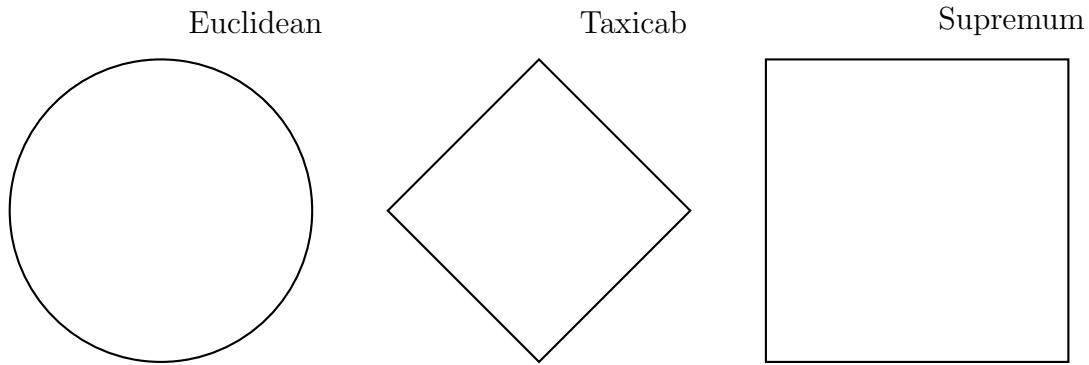
This is an **axis-aligned square** with vertices at $(1, 1)$, $(-1, 1)$, $(-1, -1)$, and $(1, -1)$.

Why? The condition $\max\{|x|, |y|\} < 1$ means both $|x| < 1$ and $|y| < 1$, which is exactly the open square $(-1, 1) \times (-1, 1)$.

(d) **Discrete Metric**:

$$B_1(0) = \begin{cases} \{(0, 0)\} & \text{if } 0 < r \leq 1, \\ \mathbb{R}^2 & \text{if } r > 1. \end{cases}$$

With radius exactly 1, only the center itself is included. With radius greater than 1, the entire space is the unit ball.



Remark 1.3 (The Shape Matters). The shape of the unit ball encodes the geometry of the metric. These three metrics are all valid, but they measure "distance" differently. A point $(0.8, 0.6)$ has:

- Euclidean norm: $\sqrt{0.64 + 0.36} = 1$ (on the circle).
- Taxicab norm: $0.8 + 0.6 = 1.4 > 1$ (outside the diamond).
- Supremum norm: $\max(0.8, 0.6) = 0.8 < 1$ (inside the square).

So the same point is on the boundary of the Euclidean unit ball, outside the taxicab ball, and inside the supremum ball!

1.5 Containment Relations of ℓ^p Spaces

For $1 \leq p \leq \infty$, recall the sequence space:

$$\ell^p = \left\{ (x_n)_{n=1}^\infty : \sum_{n=1}^\infty |x_n|^p < \infty \right\}.$$

For $p = \infty$, ℓ^∞ is the space of bounded sequences:

$$\ell^\infty = \left\{ (x_n)_{n=1}^\infty : \sup_{n \in \mathbb{N}} |x_n| < \infty \right\}.$$

A fundamental and highly useful fact is that these spaces are **nested** (contained in each other). Specifically, if $1 \leq p < q \leq \infty$, then

$$\ell^p \subset \ell^q.$$

Thus we have the chain:

$$\ell^1 \subset \ell^2 \subset \ell^3 \subset \dots \subset \ell^\infty.$$

Proof. Let $x = (x_n) \in \ell^p$. Since $\sum_{n=1}^{\infty} |x_n|^p < \infty$, we must have $x_n \rightarrow 0$ as $n \rightarrow \infty$ (the terms of a convergent series tend to zero).

Thus, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n| \leq 1$.

Since $q > p$, for $|x_n| \leq 1$, we have $|x_n|^q \leq |x_n|^p$.

Therefore,

$$\sum_{n=1}^{\infty} |x_n|^q = \sum_{n=1}^{N-1} |x_n|^q + \sum_{n=N}^{\infty} |x_n|^q \leq \sum_{n=1}^{N-1} |x_n|^q + \sum_{n=N}^{\infty} |x_n|^p < \infty.$$

Hence $x \in \ell^q$. This proves the containment. $\square$

Strict Containment

To see that the containment is **strict** (i.e., $\ell^p \neq \ell^q$), consider the sequence $x = (x_n)$ defined by

$$x_n = \frac{1}{n^{1/p}}, \quad n = 1, 2, 3, \dots$$

Then

$$\sum_{n=1}^{\infty} |x_n|^p = \sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

so $x \notin \ell^p$.

However, if $q > p$, then $q/p > 1$. The series

$$\sum_{n=1}^{\infty} |x_n|^q = \sum_{n=1}^{\infty} \frac{1}{n^{q/p}}$$

converges because it is a p -series with exponent $q/p > 1$. Therefore, $x \in \ell^q$.

This shows that ℓ^p is a **proper subset** of ℓ^q :

$$\ell^p \subsetneq \ell^q \quad \text{for } 1 \leq p < q \leq \infty.$$

Example 1.10 (Concrete Example for $\ell^1 \subset \ell^2$). The sequence $x_n = \frac{1}{n}$ satisfies:

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty \quad \Rightarrow \quad x \notin \ell^1,$$

but

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty \quad \Rightarrow \quad x \in \ell^2.$$

So ℓ^1 is strictly smaller than ℓ^2 .

Intuition and Summary

The nesting $\ell^1 \subset \ell^2 \subset \ell^3 \subset \dots \subset \ell^\infty$ tells us that:

- **Higher p means easier to be in the space.** Summability with a smaller exponent ($p = 1$) is stricter than summability with a larger exponent ($p = 2$).
- The **supremum norm** ℓ^∞ is the largest of these spaces: it only requires boundedness, which is a much weaker condition than absolute summability.

The reverse containment ($\ell^q \subset \ell^p$ for $q > p$) is **false** on $\mathbb{N}$ with counting measure. In fact, the example above shows that $\ell^q \setminus \ell^p$ is always non-empty for $p < q$. This is a key distinction between sequence spaces ℓ^p and function spaces L^p on finite measure spaces (where the containment is reversed).

2 Open and Closed Balls; Neighbourhoods

2.1 Open Balls and Closed Balls

Definition 2.1 (Open Ball). Let (X, d) be a metric space, $x \in X$, and $r > 0$. The **open ball** centered at x with radius r is:

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

Definition 2.2 (Closed Ball). The **closed ball** centered at x with radius r is:

$$\overline{B}_r(x) = \{y \in X : d(x, y) \leq r\}.$$

Remark 2.1. The notation $B_r(x)$ and $\overline{B}_r(x)$ is standard. Some authors use $\bar{B}_r(x)$ or $D_r(x)$ for the closed ball.

2.2 Examples of Open and Closed Balls

Example 2.1 (Balls in $\mathbb{R}$ with Euclidean Metric). In $\mathbb{R}$ with $d(x, y) = |x - y|$:

- $B_r(x) = (x - r, x + r)$, an open interval.
- $\overline{B}_r(x) = [x - r, x + r]$, a closed interval.

Example 2.2 (Balls in $\mathbb{R}^2$ with Euclidean Metric). In $\mathbb{R}^2$ with d_2 :

- $B_r(x_0, y_0)$ is the open disk centered at (x_0, y_0) with radius r .
- $\overline{B}_r(x_0, y_0)$ is the closed disk (including the boundary circle).

Example 2.3 (Balls in the Discrete Metric). In a discrete metric space:

- If $0 < r \leq 1$, then $B_r(x) = \{x\}$ (just the center).
- If $r > 1$, then $B_r(x) = X$ (the whole space).
- Similarly, $\overline{B}_r(x) = \{x\}$ if $r < 1$, and $\overline{B}_r(x) = X$ if $r \geq 1$.

Example 2.4 (Balls in the Taxicab Metric). In $\mathbb{R}^2$ with the taxicab metric d_1 :

$$B_r(x_0, y_0) = \{(x, y) : |x - x_0| + |y - y_0| < r\}.$$

This is an open diamond (rotated square). The boundary is given by $|x - x_0| + |y - y_0| = r$.

Example 2.5 (Balls in the Supremum Metric). In $\mathbb{R}^2$ with the supremum metric d_∞ :

$$B_r(x_0, y_0) = \{(x, y) : \max\{|x - x_0|, |y - y_0|\} < r\} = (x_0 - r, x_0 + r) \times (y_0 - r, y_0 + r).$$

This is an open axis-aligned square.

Example 2.6 (Balls in Matrix Space with Frobenius Metric). Let $M_{2 \times 2}(\mathbb{R})$ with the Frobenius metric. For the zero matrix 0 , the open ball $B_r(0)$ consists of all matrices whose Frobenius norm is less than r :

$$B_r(0) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \sqrt{a^2 + b^2 + c^2 + d^2} < r \right\}.$$

This is a 4-dimensional "ball" in the space of matrices.

Example 2.7 (Balls in Function Space with Supremum Metric). Let $X = C([0, 1])$ with $d_\infty(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$. Then:

$$B_r(f) = \{g \in C([0, 1]) : \sup_{x \in [0, 1]} |g(x) - f(x)| < r\}.$$

This consists of all continuous functions whose graphs lie strictly within a "tube" of radius r around the graph of f .

2.3 Neighbourhoods

Definition 2.3 (Neighbourhood). A **neighbourhood** of a point $x \in X$ is any set $N \subset X$ that contains an open ball centered at x .

Equivalently: N is a neighbourhood of x if there exists $r > 0$ such that $B_r(x) \subset N$.

Remark 2.2. A neighbourhood does not have to be open itself. Any set that contains an open set around x qualifies. In many advanced texts, "neighbourhood" is often reserved for open sets, but in this course we use the more general definition.

Example 2.8. In $\mathbb{R}$, the set $[0, 2]$ is a neighbourhood of 1 because $B_{1/2}(1) = (1/2, 3/2) \subset [0, 2]$. However, $[0, 2]$ is not a neighbourhood of 0 because no open ball around 0 is contained in $[0, 2]$.

3 Open Sets

3.1 Definition of Open Sets

Definition 3.1 (Open Set). A subset $U \subset X$ is **open** in the metric space (X, d) if for every point $x \in U$, there exists $r > 0$ such that $B_r(x) \subset U$.

Remark 3.1. *Intuition:* A set is open if every point in it has some "breathing room"—you can move a little bit in any direction and still stay inside the set.

Remark 3.2 (Empty Set and Whole Space). Both $\emptyset$ and X are open sets (vacuously for $\emptyset$, and trivially for X).

Example 3.1 (Open Sets in $\mathbb{R}$). In $\mathbb{R}$ with the usual metric:

- Intervals of the form (a, b) , $(-\infty, b)$, (a, ∞) , and $\mathbb{R}$ itself are open.
- Intervals of the form $[a, b]$, $(a, b]$, $[a, b)$ are **not** open.
- Singletons $\{a\}$ are not open.
- $\mathbb{Q}$ is not open (since any interval around a rational contains irrationals).

Example 3.2 (Open Sets in a Discrete Metric Space). In a discrete metric space, **every** subset is open. Why? For any $x \in A$, $B_1(x) = \{x\} \subset A$. So every point has a ball around it contained in the set.

3.2 Open Balls are Open Sets

Theorem 3.1. Every open ball $B_r(x)$ in a metric space is an open set.

Proof. Let $y \in B_r(x)$. Then $d(x, y) = t < r$. Choose $s = r - t > 0$. For any $z \in B_s(y)$, we have:

$$d(x, z) \leq d(x, y) + d(y, z) < t + s = t + (r - t) = r.$$

Thus $z \in B_r(x)$. Hence $B_s(y) \subset B_r(x)$. So every point of $B_r(x)$ has a ball around it contained in $B_r(x)$. Therefore $B_r(x)$ is open. $\square$

3.3 Properties of Open Sets

Theorem 3.2 (Arbitrary Unions). The union of any collection of open sets is open.

Proof. Let $\{U_\alpha\}_{\alpha \in I}$ be a collection of open sets, and let $U = \bigcup_{\alpha \in I} U_\alpha$. Take $x \in U$. Then $x \in U_\alpha$ for some α . Since U_α is open, there exists $r > 0$ such that $B_r(x) \subset U_\alpha \subset U$. Hence U is open. $\square$

Theorem 3.3 (Finite Intersections). The intersection of finitely many open sets is open.

Proof. Let $U_1, \dots, U_n$ be open. Let $U = \bigcap_{k=1}^n U_k$. Take $x \in U$. For each k , there exists $r_k > 0$ such that $B_{r_k}(x) \subset U_k$. Let $r = \min\{r_1, \dots, r_n\} > 0$. Then $B_r(x) \subset U_k$ for all k , so $B_r(x) \subset U$. Hence U is open. $\square$

Remark 3.3 (Infinite Intersections). The intersection of infinitely many open sets need not be open. For example, in $\mathbb{R}$:

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\},$$

and $\{0\}$ is not open in $\mathbb{R}$.

3.4 Characterization of Open Sets

Theorem 3.4 (Union of Open Balls). A subset $U \subset X$ is open if and only if it can be expressed as a union of open balls.

Proof. ($\Rightarrow$): If U is open, then for each $x \in U$, choose $r_x > 0$ such that $B_{r_x}(x) \subset U$. Then $U = \bigcup_{x \in U} B_{r_x}(x)$.

($\Leftarrow$): A union of open balls is open because each ball is open and arbitrary unions of open sets are open. $\square$

Remark 3.4. This theorem shows that open sets in a metric space are precisely the sets that can be "built" from open balls.

4 Closed Sets and Limit Points

4.1 Definition of Closed Sets

Definition 4.1 (Closed Set). A subset $F \subset X$ is **closed** if its complement $X \setminus F$ is open.

Remark 4.1 (Clopen Sets). Some sets can be both open and closed (e.g., $\emptyset$ and X in any metric space). These are called **clopen** sets. In a discrete metric space, **every** subset is clopen.

4.2 Limit Points (Accumulation Points)

Definition 4.2 (Limit Point). A point $x \in X$ is a **limit point** (or **accumulation point** or **cluster point**) of a set $A \subset X$ if every neighbourhood of x contains a point of A different from x .

Equivalently: for every $r > 0$, $B_r(x) \cap (A \setminus \{x\}) \neq \emptyset$.

Remark 4.2 (Intuition). A point x is a limit point of A if there are points of A arbitrarily close to x , but not necessarily equal to x . The point itself need not belong to A .

Example 4.1 (Limit Points in $\mathbb{R}$). • The limit points of $(0, 1)$ are $[0, 1]$.

- The limit points of $\mathbb{Q}$ are all of $\mathbb{R}$.
- The limit points of $\mathbb{Z}$ are $\emptyset$ (no limit points).
- The limit points of $\{\frac{1}{n} : n \in \mathbb{N}\}$ is $\{0\}$.

Example 4.2 (Limit Points in a Discrete Metric Space). In a discrete metric space, no set has any limit points. For any $x \in X$, $B_1(x) = \{x\}$, so no other point of A can be in the ball.

4.3 Isolated Points

Definition 4.3 (Isolated Point). A point $x \in A$ is an **isolated point** of A if it is **not** a limit point of A .

Equivalently: there exists $r > 0$ such that $B_r(x) \cap A = \{x\}$.

Remark 4.3. Every point of a set is either a limit point or an isolated point (or both, if it's a limit point that also belongs to the set and has other points arbitrarily close).

4.4 Characterization of Closed Sets via Limit Points

Theorem 4.1 (Closed Sets Contain Their Limit Points). A set $F \subset X$ is closed if and only if it contains all its limit points.

Proof. ($\Rightarrow$): Suppose F is closed, so F^c is open. Let x be a limit point of F . If $x \notin F$, then $x \in F^c$. Since F^c is open, there exists $r > 0$ with $B_r(x) \subset F^c$. Then $B_r(x) \cap F = \emptyset$, contradicting that x is a limit point of F . Hence $x \in F$.

($\Leftarrow$): Suppose F contains all its limit points. We show F^c is open. Take $x \in F^c$. Since x is not a limit point of F , there exists $r > 0$ such that $B_r(x) \cap F = \emptyset$. Thus $B_r(x) \subset F^c$. Hence F^c is open, so F is closed. $\square$

4.5 Properties of Closed Sets

Theorem 4.2 (Arbitrary Intersections). The intersection of any collection of closed sets is closed.

Proof. Let $\{F_\alpha\}$ be closed sets. Then $X \setminus F_\alpha$ is open. By De Morgan's laws:

$$X \setminus \left(\bigcap_{\alpha} F_{\alpha} \right) = \bigcup_{\alpha} (X \setminus F_{\alpha}),$$

which is open (union of open sets). Hence $\bigcap_{\alpha} F_{\alpha}$ is closed. $\square$

Theorem 4.3 (Finite Unions). The union of finitely many closed sets is closed.

Proof. Similar to the above, using De Morgan's laws and the fact that finite intersections of open sets are open. $\square$

Remark 4.4 (Infinite Unions). The union of infinitely many closed sets need not be closed. For example, in $\mathbb{R}$:

$$\bigcup_{n=1}^{\infty} \left[\frac{1}{n}, 1 \right] = (0, 1],$$

which is not closed (since 0 is a limit point not in the set).

5 Interior, Closure, and Boundary

5.1 Interior of a Set

Definition 5.1 (Interior Point). A point $x \in X$ is an **interior point** of a set $A \subset X$ if there exists $r > 0$ such that $B_r(x) \subset A$.

Definition 5.2 (Interior). The **interior** of A , denoted $\text{int}(A)$ or A° , is the set of all interior points of A .

Theorem 5.1 (Properties of Interior). (i) $\text{int}(A)$ is open.

(ii) $\text{int}(A) \subset A$.

(iii) $\text{int}(A)$ is the largest open set contained in A .

(iv) A is open if and only if $A = \text{int}(A)$.

Example 5.1 (Interior in $\mathbb{R}$). • $\text{int}((0, 1)) = (0, 1)$

- $\text{int}([0, 1]) = (0, 1)$
- $\text{int}(\mathbb{Q}) = \emptyset$
- $\text{int}(\mathbb{Z}) = \emptyset$
- $\text{int}((0, 1] \cup \{2\}) = (0, 1)$

5.2 Closure of a Set

Definition 5.3 (Closure). The **closure** of a set $A \subset X$, denoted $\overline{A}$, is the intersection of all closed sets containing A .

Theorem 5.2 (Characterization of Closure).

$$\overline{A} = A \cup \{\text{limit points of } A\}.$$

Theorem 5.3 (Properties of Closure). (i) $\overline{A}$ is closed.

(ii) $A \subset \overline{A}$.

(iii) $\overline{A}$ is the smallest closed set containing A .

(iv) A is closed if and only if $A = \overline{A}$.

(v) $\overline{\overline{A}} = \overline{A}$ (idempotence).

Example 5.2 (Closure in $\mathbb{R}$). • $\overline{(0, 1)} = [0, 1]$

- $\overline{[0, 1]} = [0, 1]$
- $\overline{\mathbb{Q}} = \mathbb{R}$
- $\overline{\mathbb{Z}} = \mathbb{Z}$
- $\overline{\left\{\frac{1}{n} : n \in \mathbb{N}\right\}} = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\}$

Example 5.3 (Closure in Matrix Space). Let $GL(2, \mathbb{R})$ be the set of invertible 2×2 real matrices. Its closure in $M_{2 \times 2}(\mathbb{R})$ with the Frobenius metric is the set of all 2×2 matrices, since the determinant function is continuous and matrices with determinant 0 are limit points of invertible matrices.

5.3 Boundary Points

Definition 5.4 (Boundary Point). A point $x \in X$ is a **boundary point** of a set $A \subset X$ if every neighbourhood of x intersects both A and its complement $X \setminus A$.

Definition 5.5 (Boundary). The **boundary** of A , denoted ∂A , is the set of all boundary points of A .

Theorem 5.4 (Boundary Characterization).

$$\partial A = \overline{A} \setminus \text{int}(A).$$

Example 5.4 (Boundary in $\mathbb{R}$). • $\partial(0, 1) = \{0, 1\}$

- $\partial[0, 1] = \{0, 1\}$
- $\partial\mathbb{Q} = \mathbb{R}$ (every point is arbitrarily close to both rationals and irrationals)
- $\partial\mathbb{Z} = \mathbb{Z}$ (every integer has rationals on both sides)
- $\partial\left\{\frac{1}{n} : n \in \mathbb{N}\right\} = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\}$

Example 5.5 (Boundary in a Discrete Metric Space). For any set A in a discrete metric space, $\partial A = \emptyset$. Why? Every point has a neighbourhood $(\{x\})$ that is either entirely in A or entirely in A^c . So no point is a boundary point.

5.4 Exterior Points

Definition 5.6 (Exterior Point). A point $x \in X$ is an **exterior point** of A if it is an interior point of the complement $X \setminus A$.

Definition 5.7 (Exterior). The **exterior** of A , denoted $\text{ext}(A)$, is $\text{int}(X \setminus A)$.

Remark 5.1. The whole space X is partitioned into three disjoint sets:

$$X = \text{int}(A) \cup \partial A \cup \text{ext}(A).$$

5.5 Summary of Relationships

For any set $A \subset X$:

- $\text{int}(A) \subset A \subset \overline{A}$
- A is open $\iff A = \text{int}(A)$
- A is closed $\iff A = \overline{A}$
- $\partial A = \overline{A} \setminus \text{int}(A)$
- $\overline{A} = A \cup \partial A$
- $\partial A = \partial(X \setminus A)$

6 Bounded Sets and Diameter

6.1 Bounded Sets

Definition 6.1 (Bounded Set). A subset $A \subset X$ of a metric space is **bounded** if it is contained in some open ball $B_r(x)$.

Equivalently: there exists $x \in X$ and $r > 0$ such that $A \subset B_r(x)$.

Example 6.1 (Bounded Sets in $\mathbb{R}$). In $\mathbb{R}$ with the usual metric, bounded sets are exactly those that are contained in some interval (a, b) .

- $(0, 1)$, $[0, 1]$, and finite sets are bounded.
- $\mathbb{R}$, $(0, \infty)$, and $\mathbb{Z}$ are unbounded.

Example 6.2 (Bounded Sets in a Discrete Metric Space). In a discrete metric space, **every** set is bounded. For any set A , choose any $x \in A$ and $r = 2$. Then $A \subset B_2(x) = X$.

Example 6.3 (Bounded Sets in Function Spaces). In $C([0, 1])$ with the supremum metric, a set $\mathcal{F}$ is bounded if there exists $M > 0$ such that $\|f\|_\infty \leq M$ for all $f \in \mathcal{F}$. That is, all functions in the set are uniformly bounded by the same constant M .

6.2 Diameter of a Set

Definition 6.2 (Diameter). The **diameter** of a non-empty set $A \subset X$ is:

$$\text{diam}(A) = \sup\{d(x, y) : x, y \in A\}.$$

If A is empty, we define $\text{diam}(\emptyset) = 0$.

Theorem 6.1 (Boundedness Criterion). A non-empty set A is bounded if and only if $\text{diam}(A) < \infty$.

Proof. ($\Rightarrow$): If $A \subset B_r(x)$, then for all $a, b \in A$, $d(a, b) \leq d(a, x) + d(x, b) < 2r$. So $\text{diam}(A) \leq 2r < \infty$.

($\Leftarrow$): If $\text{diam}(A) = D < \infty$, fix any $x \in A$. Then for all $a \in A$, $d(x, a) \leq D$. Thus $A \subset B_{D+1}(x)$, so A is bounded. $\square$

Example 6.4 (Diameter in $\mathbb{R}$). • $\text{diam}((0, 1)) = 1$

- $\text{diam}([0, 1]) = 1$
- $\text{diam}(\mathbb{Q} \cap (0, 1)) = 1$
- $\text{diam}(\mathbb{N}) = \infty$ (unbounded)

Example 6.5 (Diameter in the Discrete Metric). In a discrete metric space, for any set A :

$$\text{diam}(A) = \begin{cases} 0 & \text{if } |A| \leq 1, \\ 1 & \text{if } |A| \geq 2. \end{cases}$$

So every non-empty finite set with at least two points has diameter 1.

6.3 Distance Between Sets

Definition 6.3 (Distance from a Point to a Set). For a point $x \in X$ and a non-empty set $A \subset X$, define:

$$d(x, A) = \inf\{d(x, a) : a \in A\}.$$

Definition 6.4 (Distance Between Sets). For non-empty sets $A, B \subset X$, define:

$$d(A, B) = \inf\{d(a, b) : a \in A, b \in B\}.$$

Remark 6.1. The distance between sets is **not** a metric on subsets. It is symmetric and non-negative, but $d(A, B) = 0$ does not imply $A = B$ (e.g., in $\mathbb{R}$, $d((0, 1), (1, 2)) = 0$ but the sets are disjoint).

Example 6.6 (Distance in $\mathbb{R}$). • $d(0, [1, 2]) = 1$

- $d(0, (0, 1)) = 0$
- $d([0, 1], [2, 3]) = 1$
- $d((0, 1), (1, 2)) = 0$

Theorem 6.2 (Distance to a Closed Set). If A is closed and $x \notin A$, then $d(x, A) > 0$.

Proof. If $d(x, A) = 0$, then for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $d(x, a_n) < 1/n$. Then $a_n \rightarrow x$, so x is a limit point of A . Since A is closed, $x \in A$, contradicting $x \notin A$. Hence $d(x, A) > 0$. $\square$

7 Subspace of a Metric Space

7.1 The Subspace Metric

Definition 7.1 (Subspace). Let (X, d) be a metric space and let $Y \subset X$ be any non-empty subset. Then $(Y, d|_{Y \times Y})$ is a metric space, called a **subspace** of X .

The metric $d|_{Y \times Y}$ is the restriction of d to pairs of points in Y .

7.2 Relative Open and Closed Sets

Definition 7.2 (Relatively Open). A set $E \subset Y$ is **relatively open** in Y (or open in the subspace Y) if it is open in the metric space $(Y, d|_{Y \times Y})$.

Theorem 7.1 (Characterization of Relatively Open Sets). A set $E \subset Y$ is relatively open in Y if and only if there exists an open set $U \subset X$ such that $E = U \cap Y$.

Proof. ($\Rightarrow$): If E is open in Y , for each $y \in E$ there exists $r_y > 0$ such that $B_{r_y}^Y(y) \subset E$. Let $U = \bigcup_{y \in E} B_{r_y}^X(y)$. Then U is open in X and $U \cap Y = E$.

($\Leftarrow$): If $E = U \cap Y$ with U open in X , then for any $y \in E$, there exists $r > 0$ such that $B_r^X(y) \subset U$. Then $B_r^Y(y) = B_r^X(y) \cap Y \subset E$. Hence E is open in Y . $\square$

Example 7.1 (Subspace Example). In $X = \mathbb{R}$ with the usual metric, let $Y = [0, 1]$. Then:

- $E = [0, 1/2)$ is relatively open in Y because $E = Y \cap (-1, 1/2)$, and $(-1, 1/2)$ is open in $\mathbb{R}$.
- However, E is **not** open in $\mathbb{R}$ (since 0 is in E but no full interval around 0 is contained in E).
- Similarly, $E = (1/4, 3/4]$ is relatively open in Y because $E = Y \cap (1/4, \infty)$.

Example 7.2 (Relatively Closed Sets). A set $E \subset Y$ is relatively closed in Y iff $E = F \cap Y$ for some closed set $F \subset X$.

In $Y = [0, 1]$, the set $E = [0, 1/2]$ is relatively closed in Y because $E = Y \cap [0, 1/2]$, and $[0, 1/2]$ is closed in $\mathbb{R}$.

8 Tutorial / Problem-Solving Session

8.1 Key Theorems to Master

1. **Triangle Inequality Variant:** $|d(x, z) - d(y, z)| \leq d(x, y)$.
2. **Open Ball is Open:** Every open ball is an open set.
3. **Union of Open Sets:** Arbitrary unions of open sets are open.
4. **Finite Intersection of Open Sets:** Finite intersections of open sets are open.
5. **Closed iff Contains Limit Points:** A set is closed iff it contains all its limit points.
6. **Closure Characterization:** $\overline{A} = A \cup A'$ (where A' is the set of limit points).
7. **Boundary Formula:** $\partial A = \overline{A} \setminus \text{int}(A)$.
8. **Subspace Characterization:** E is open in Y iff $E = U \cap Y$ for some open $U \subset X$.

8.2 Worked Examples

Example 8.1 (Interior, Closure, Boundary in $\mathbb{R}$). Find the interior, closure, and boundary of the following sets in $\mathbb{R}$ with the usual metric:

1. $A = (0, 1] \cup \{2\}$
 - $\text{int}(A) = (0, 1)$
 - $\overline{A} = [0, 1] \cup \{2\}$
 - $\partial A = \{0, 1, 2\}$
2. $B = \mathbb{Q}$
 - $\text{int}(B) = \emptyset$
 - $\overline{B} = \mathbb{R}$
 - $\partial B = \mathbb{R}$

3. $C = \mathbb{Z}$
- $\text{int}(C) = \emptyset$
- $\overline{C} = \mathbb{Z}$
- $\partial C = \mathbb{Z}$

Example 8.2 (Limit Points). Find all limit points of the following sets:

1. $A = \{1/n : n \in \mathbb{N}\}$
 - Limit point: 0 only.
 - Every point of A is isolated.
2. $B = \mathbb{Q} \cap (0, 1)$
 - Limit points: $[0, 1]$.
3. $C = \mathbb{N}$
 - No limit points (all points are isolated).

Example 8.3 (Subspace Topology). Let $X = \mathbb{R}$ with the usual metric, and let $Y = [0, 1]$. Determine whether the following sets are open in Y :

1. $E_1 = [0, 1/2)$: Open in Y because $E_1 = Y \cap (-1, 1/2)$.
2. $E_2 = [0, 1/2]$: Not open in Y because no neighbourhood of $1/2$ in Y is contained in E_2 .
3. $E_3 = (1/4, 3/4)$: Open in Y because $E_3 = Y \cap (1/4, 3/4)$.

8.3 Practice Problems

1. Let (X, d) be a metric space. Prove that the union of any collection of open sets is open.
2. Prove that a finite intersection of open sets is open. Give an example showing that infinite intersections need not be open.
3. Show that in a metric space, every closed ball $\overline{B}_r(x) = \{y : d(x, y) \leq r\}$ is a closed set.
4. Let $A \subset X$. Prove that $\partial A = \overline{A} \setminus \text{int}(A)$.
5. In $\mathbb{R}$ with the usual metric, find the interior, closure, and boundary of:
 - (a) $(0, 1) \cup (1, 2)$
 - (b) $[0, 1) \cup \{2\}$
 - (c) $\mathbb{R} \setminus \mathbb{Q}$
6. Let X be a discrete metric space. Describe all open sets, closed sets, limit points, and isolated points.
7. Prove that in any metric space, a set A is closed if and only if it contains all its limit points.

8. Let Y be a subspace of a metric space X . Prove that a set $E \subset Y$ is relatively open in Y iff $E = U \cap Y$ for some open set $U \subset X$.
9. Compute the diameter of the following sets in $\mathbb{R}$:
- (a) $(0, 1) \cup (1, 2)$
 - (b) $\mathbb{Q} \cap [0, 1]$
 - (c) $\{1, 2, 3, \dots\}$
10. In $\mathbb{R}^2$ with the taxicab metric, describe the open ball $B_1(0, 0)$ and sketch its shape.

Summary Table: Key Concepts in Unit II

Concept	Definition	Key Property
Metric	$d : X \times X \rightarrow [0, \infty)$ satisfying positivity, symmetry, triangle inequality	Abstracts "distance"
Open Ball	$B_r(x) = \{y : d(x, y) < r\}$	Always an open set
Closed Ball	$\overline{B}_r(x) = \{y : d(x, y) \leq r\}$	Always a closed set
Open Set	Every point has a ball around it contained in the set	Arbitrary unions, finite intersections
Closed Set	Complement is open	Contains all limit points
Limit Point	Every neighbourhood contains a point of the set different from it	Characterizes closed sets
Isolated Point	Point in set that is not a limit point	Has a neighbourhood containing only that point
Interior	$\text{int}(A) = \{x : \exists r > 0, B_r(x) \subset A\}$	Largest open subset of A
Closure	$\overline{A} = A \cup \{\text{limit points}\}$	Smallest closed set containing A
Boundary	$\partial A = \overline{A} \setminus \text{int}(A)$	Points arbitrarily close to both A and A^c
Bounded Set	Contained in some ball	Finite diameter
Diameter	$\text{diam}(A) = \sup\{d(x, y) : x, y \in A\}$	Measures "size" of set
Subspace	$Y \subset X$ with restricted metric	Relative open/closed sets

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